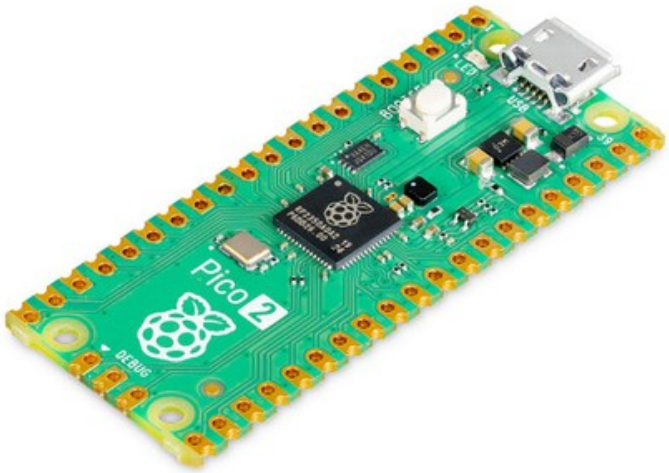
**********Draft**********

High Speed Data Acquisition Over HDMI hsdaoh

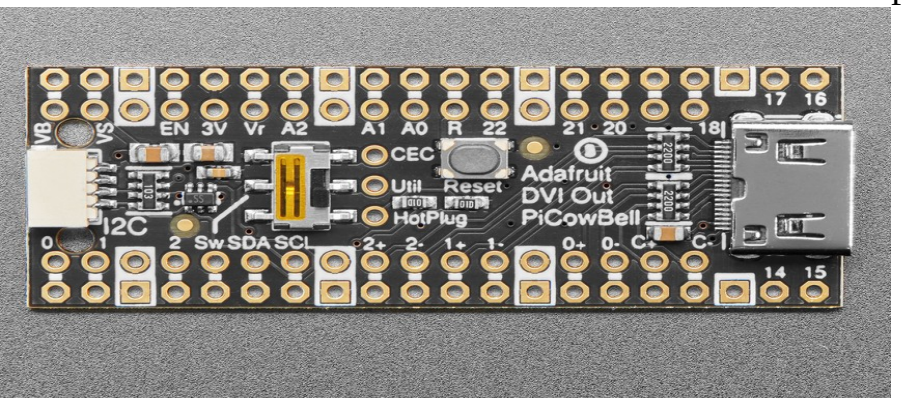
10/30/26

**********Draft**********

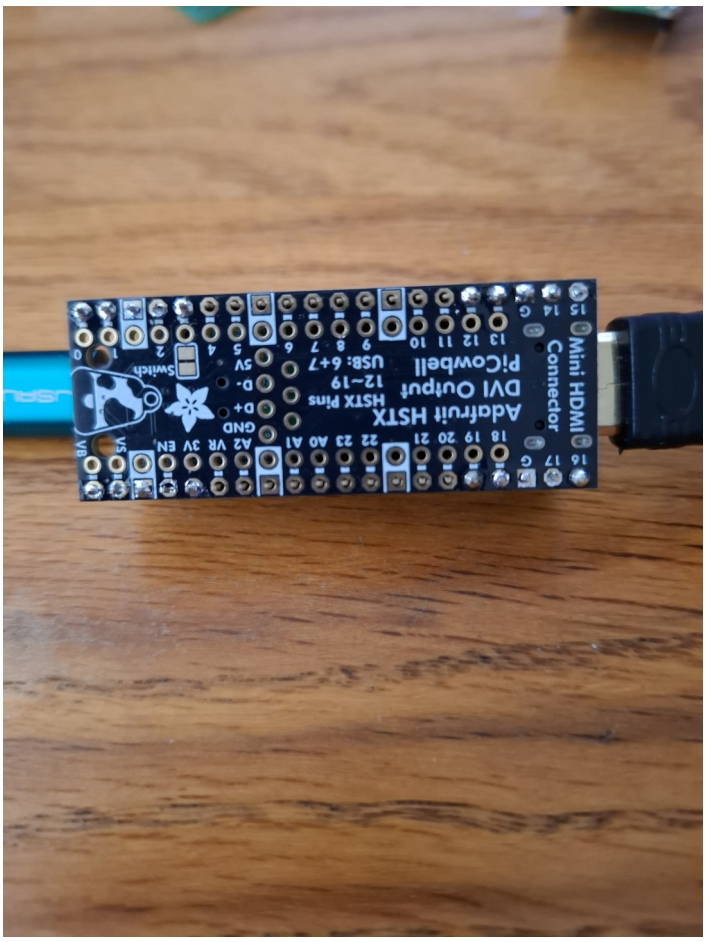
pico2



PicowBell-dvi



xx



Test pico2 with PicowBell-dvi

https://github.com/develone/pico-examples/blob/test-dev/doc/hstx/mtn-hstx-wkg-091426.jpg

https://github.com/develone/pico-examples/blob/test-dev/doc/hstx/lena_640x480_rgb332.h.jpg

The information below was found at “https://www.electronics-lab.com/news/the-ms2130-mini-featuring-the-raspberry-pi-pico-2s-hstx-interface-is-an-hdmi-to-usb-3-0-4k-video-capture-card-that-supports-1080p-60fps-for-live-streaming-and-recording/”
xx

4K Video Capture Card, Cam Link Card Game Capture Card Audio Capture Adapter HDMI to USB 2.0 Record Capture Device for Streaming, Live Broadcasting, Video Conference, Teaching, Gaming(Black)

4.3*4.3 out of 5 stars*
(3.2K)
100+ bought in past month
Price, product page$9.98



xx

The MS2130 Mini, featuring the Raspberry Pi Pico 2's HSTX interface, is an HDMI to USB 3.0 4K video capture card that supports 1080p 60fps for live streaming and recording.

The MS2130 Mini 4K Video Capture Card is a compact and high-quality device designed for capturing HDMI video and audio signals for use on computers or online streaming platforms. It supports input resolutions up to 4K@30Hz and outputs at 1080P@60fps with YUY2 full HD color depth, ensuring sharp, detailed visuals.

The above site also had a link to "https://github.com/steve-m/hsdaoh-rp2350" for the Raspberry Pi pico2.

To build hsdaoh-rp2350: After creating a fork of the repo and the test-dev branch.

```
devel@pi5-80:~/hsdaoh-rp2350/build $ git clone -b test-dev
https://github.com/develone/hsdaoh-rp2350.git

devel@pi5-80:~/hsdaoh-rp2350/build $ mkdir hsdaoh-rp2350/build

devel@pi5-80:~/hsdaoh-rp2350/build $ cd hsdaoh-rp2350/build

devel@pi5-80:~/hsdaoh-rp2350/build $ cmake -DPICO_PLATFORM=rp2350 -
DPICO_BOARD=pico2 ../
```

```
devel@pi5-80:~/hsdaoh-rp2350/build $ make -j 8
```

```
devel@pi5-80:~/hsdaoh-rp2350/build $ ls apps/counter/
CMakeFiles          counter.bin  counter.elf      counter.hex    counter.uf2
cmake_install.cmake  counter.dis  counter.elf.map  counter.pio.h  Makefile
```

needed additional packages

```
devel@pi5-90:~/hsdaoh/build $  sudo apt install libflac
devel@pi5-90:~/hsdaoh/build $ sudo apt install flac
devel@pi5-90:~/hsdaoh/build $ sudo ldconfig
devel@pi5-90:~/hsdaoh/build $  sudo apt install libflac
devel@pi5-90:~/hsdaoh/build $ s sudo apt install libflac-dev
devel@pi5-90:~/hsdaoh/build $ sudo apt install -y libuvc-dev v4l-utils guvcview
```

```
 devel@pi5-90:~/hsdaoh/build $ git clone https://github.com/steve-m/hsdaoh.git
 devel@pi5-90:~/hsdaoh/build $ mkdir hsdaoh/build
 devel@pi5-90:~/hsdaoh/build $ cd hsdaoh/build
 devel@pi5-90:~/hsdaoh/build $ cmake ../ -DINSTALL_UDEV_RULES=ON
 devel@pi5-90:~/hsdaoh/build $ smake -j 4
```

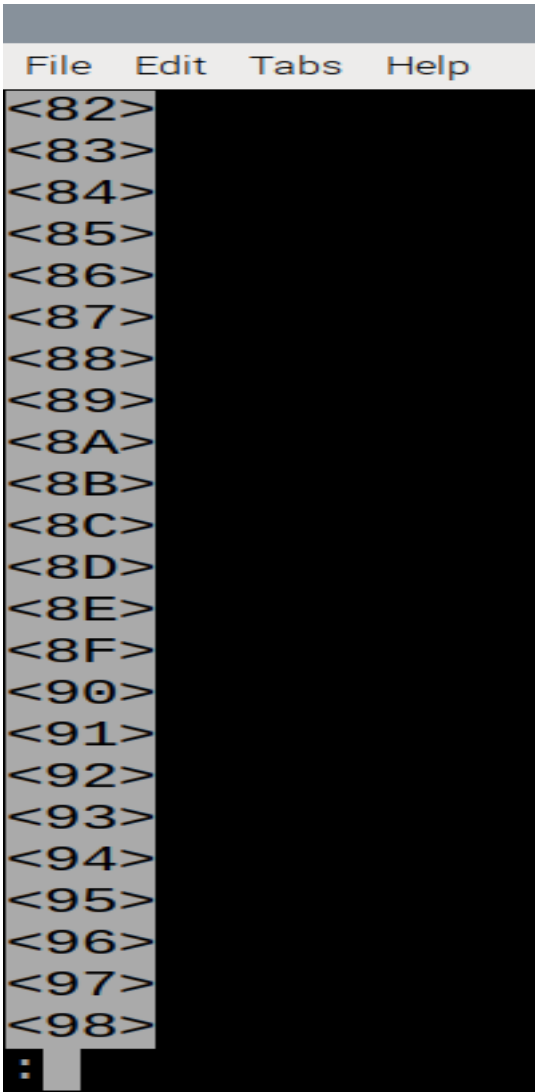
```
devel@pi5-90:~/hsdaoh/build $ sudo make install
-- Install configuration: "Release"
-- Installing: /etc/udev/rules.d/71-hsdaoh.rules
-- Installing: /usr/local/include/hsdaoh.h
-- Installing: /usr/local/include/hsdaoh_export.h
-- Installing: /usr/local/lib/pkgconfig/libhsdaoh.pc
-- Installing: /usr/local/lib/cmake/hsdaoh/hsdaohTargets.cmake
-- Installing: /usr/local/lib/cmake/hsdaoh/hsdaohTargets-release.cmake
-- Installing: /usr/local/lib/cmake/hsdaoh/hsdaohConfig.cmake
-- Installing: /usr/local/lib/cmake/hsdaoh/hsdaohConfigVersion.cmake
-- Installing: /usr/local/lib/libhsdaoh.so.0.1.0
-- Installing: /usr/local/lib/libhsdaoh.so.0
-- Installing: /usr/local/lib/libhsdaoh.so
-- Installing: /usr/local/lib/libhsdaoh.a
-- Installing: /usr/local/bin/hsdaoh_file
-- Set runtime path of "/usr/local/bin/hsdaoh_file" to ""
-- Installing: /usr/local/bin/hsdaoh_tcp
-- Set runtime path of "/usr/local/bin/hsdaoh_tcp" to ""
-- Installing: /usr/local/bin/hsdaoh_test
-- Set runtime path of "/usr/local/bin/hsdaoh_test" to ""
```

```
devel@pi5-90:~/hsdaoh/build $ hsdaoh_test
unsupported descriptor subtype VS_COLORFORMAT
unsupported descriptor subtype VS_COLORFORMAT
Reporting PPM error measurement every 10 seconds...
Press ^C after a few minutes.
attempt to claim already-claimed interface 1
Synchronized to HDMI input stream
real sample rate: 64023817 current PPM: 1134127 cumulative PPM: 1134127
real sample rate: 63957098 current PPM: 1131903 cumulative PPM: 1132980
```

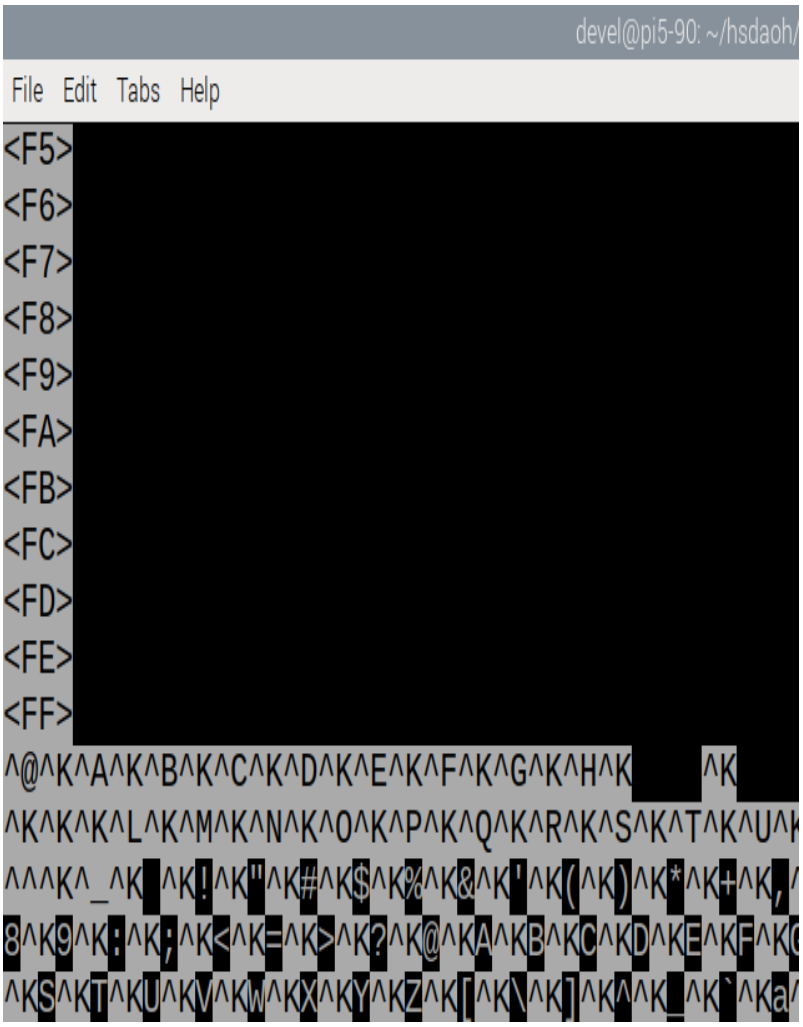
real sample rate: 64048601 current PPM: 1134953 cumulative PPM: 1133652
real sample rate: 63952631 current PPM: 1131754 cumulative PPM: 1133169
:

devel@pi5-90:~/hsdaoh $ hsdaoh_test
unsupported descriptor subtype VS_COLORFORMAT
unsupported descriptor subtype VS_COLORFORMAT
Reporting PPM error measurement every 10 seconds...
Press ^C after a few minutes.
attempt to claim already-claimed interface 1
Synchronized to HDMI input stream
real sample rate: 64004096 current PPM: 1133470 cumulative PPM: 1133470
real sample rate: 63939347 current PPM: 1131312 cumulative PPM: 1132380
real sample rate: 64064408 current PPM: 1135480 cumulative PPM: 1133419
^CSignal caught, exiting!

devel@pi5-90:~/hsdaoh $ devel@pi5-90:~/hsdaoh $ hsdaoh_file > counter.txt



xx


devel@pi5-90: ~/hsdaoh/
File Edit Tabs Help
<F5>
<F6>
<F7>
<F8>
<F9>
<FA>
<FB>
<FC>
<FD>
<FE>
<FF>
^@^K^A^K^B^K^C^K^D^K^E^K^F^K^G^K^H^K ^K
^K^K^K^L^K^M^K^N^K^O^K^P^K^Q^K^R^K^S^K^T^K^U^K
^^^K^_ ^K !^K"^K#^K$^K%^K&^K'^K(^K)^K*^K+^K,^
8^K9^K:^K;^K<^K=^K>^K?^K@^KA^KB^KC^KD^KE^KF^KG
^KS^KT^KU^KV^KW^KX^KY^KZ^K[^K\^K]^K^^K_^K`^Ka^